Enterprise HRMS Platform 2.0

Comprehensive System Architecture, End-to-End Workflow Blueprints & Tracking Matrix

Version: 2.0.0 Enterprise | **Status:** Production Ready | **Stack:** Spring Boot 3.2 + React 18 + PostgreSQL + WebRTC Mesh
Author: HRMS Architecture Committee | **Date:** November 2026

1. Executive System Architecture Overview

The Enterprise HRMS Platform 2.0 provides an enterprise-wide human resource ecosystem spanning recruitment, attendance, statutory leaves, payroll, asset lifecycle, performance appraisals, and AI workforce intelligence. The architecture is fully decoupled with strict role-based access control (RBAC) and high-concurrency P2P WebRTC media streaming.

Tier	Core Frameworks & Responsibilities
Client Tier	React 18, Vite, Redux Toolkit, Tailwind CSS, Native WebRTC MediaStream, Lucide Icons
Gateway & Security	Spring Security 6, JwtAuthenticationFilter, TenantContext, CORS Whitelisting, BCrypt
Service & Logic	22 Modular Domain Services (Auth, Employee, Recruitment, Attendance, Leave, Payroll, Performance, Assets, Expense, Offboarding, Helpdesk, Intelligence)
Persistence Tier	PostgreSQL / H2, Spring Data JPA, Hibernate, Local Document Vault / Storage

2. Detailed End-to-End Workflow Blueprints

Flow 1: Tenant Provisioning, Authentication & RBAC Resolution

Governs enterprise multi-tenant tenant isolation, login credential verification, JWT token lifecycle (15m access / 7d refresh), and dynamic role-based permission resolution.

Operational Execution Steps:

- **1. User Submission:** Employee, HR, Manager, or Candidate submits email and password via login interface.
- **2. Controller Routing:** AuthController routes payload to AuthService.authenticate().
- **3. Tenant & Account Lookup:** Database queries User entity by email with active tenant scope resolution.
- **4. Security Validation:** BCryptPasswordEncoder validates password hash; account active status is verified.
- **5. JWT Token Generation:** JwtTokenProvider creates 15-minute accessToken and 7-day secure refreshToken with role claims.
- **6. Client Bootstrap:** Frontend Redux stores token; UI resolves role permissions (Admin, HR, Manager, Employee, Candidate) and redirects.

Mermaid Diagram Specification:

```
sequenceDiagram
autonumber
actor User as Corporate User / Candidate
participant UI as React Frontend
participant AuthCtrl as AuthController
participant AuthSvc as AuthService
participant TokenProvider as JwtTokenProvider
participant DB as Database (User, Role, Tenant)

User->>UI: Enters Email & Password
UI->>AuthCtrl: POST /api/v1/auth/login { email, password }
AuthCtrl->>AuthSvc: authenticate(email, password)
AuthSvc->>DB: findByEmail(email) with Tenant lookup
DB-->>AuthSvc: UserEntity + Roles + PasswordHash
AuthSvc->>AuthSvc: Verify BCrypt Password & Account Status
AuthSvc->>TokenProvider: generateToken(UserPrincipal)
TokenProvider-->>AuthSvc: accessToken (15m) + refreshToken (7d)
AuthSvc-->>AuthCtrl: AuthResponseDto (tokens, user profile, role permissions)
AuthCtrl-->>UI: 200 OK HTTP Response
UI->>UI: Store tokens in Redux + localStorage
UI->>UI: Resolve Navigation by Role (Admin, HR, Manager, Employee)
UI-->>User: Render Dashboard
```

Flow 2: Employee Lifecycle & Onboarding Workflow

Manages employee profile creation (via manual entry or 1-click ATS candidate conversion), statutory ID verification, department hierarchy linking, and statutory leave quota provisioning.

Operational Execution Steps:

- **1. Onboarding Trigger:** Initiated either via manual HR entry or 1-click recruitment hire conversion.
- **2. Profile Ingestion:** Captures personal info, contact, emergency details, and tax/statutory numbers (PAN/Aadhaar/SSN).
- **3. Hierarchy Assignment:** Assigns Employee to Department, Designation, Work Location, Shift (GEN-01), and Reporting Manager.
- **4. Automatic Quota Provisioning:** Automatically initializes statutory leave balances: Annual (24), Sick (12), Casual (8).
- **5. Credential Generation:** Generates corporate user account with BCrypt password hash and assigns ROLE_EMPLOYEE.
- **6. Welcome Dispatch:** Dispatches automated welcome email with login credentials and self-service portal link.

Mermaid Diagram Specification:

```
flowchart TD
Start([Initiate Employee Onboarding]) --> SourceCheck{Creation Source?}
SourceCheck -->|Manual HR Entry| HRForm[HR Admin Enters Employee Details]
SourceCheck -->|ATS Hire Conversion| ATSConvert[1-Click Conversion from Candidate Record]

HRForm --> ValidateProfile[Validate Unique Email, Phone, PAN/Tax ID]
ATSConvert --> ValidateProfile

ValidateProfile --> CreateEmpRecord[Insert Employee Entity Status = ACTIVE]
CreateEmpRecord --> AssignOrg[Assign Department, Designation & Shift GEN-01]
AssignOrg --> SetManager[Link Reporting Manager Hierarchy]

SetManager --> AutoQuota[Provision Statutory Leave Quotas Annual=24, Sick=12, Casual=8]
AutoQuota --> CreateUserAcct[Generate Corporate Auth User & Password Hash]
CreateUserAcct --> DocUpload[Upload KYC Documents ID, Address, Tax Forms]
DocUpload --> EmailWelcome[Dispatch Welcome Email with Login Credentials]
EmailWelcome --> Complete([Employee Onboarding Completed])
```

Flow 3: Recruitment & End-to-End ATS Workflow

Covers job opening requisitions, candidate resume parsing, interview scheduling, tokenized guest invitations, multi-party WebRTC mesh video sessions with tie-breaker glare resolution, collaborative scoring, and hiring conversion.

Operational Execution Steps:

- **1. Requisition & Sourcing:** HR creates job posting; candidate applies with resume file.
- **2. Screening & Scheduling:** Recruiter reviews profile, sets stage to INTERVIEW_ROUND_1, and assigns interviewer panel.
- **3. Secure Invitation:** Backend generates secure guestAccessToken and dispatches calendar invitations with tokenized meeting links.
- **4. Live Room Lobby:** Candidate tests webcam/mic in lobby; interviewer accesses room through corporate portal.
- **5. Multi-Peer WebRTC Mesh:** Determines offerer via lexicographical tie-breaker (myIdentity < peerId), eliminating glare; connects 3-way split.
- **6. Scorecard & Hire Conversion:** Interviewers grade candidate live; HR completes interview and clicks 1-click candidate-to-employee conversion.

Mermaid Diagram Specification:

```
sequenceDiagram
autonumber
actor HR as HR Recruiter
actor Cand as Candidate
actor Panel as Co-Interviewer Panel
participant ATS as Recruitment & ATS Service
participant Signal as InterviewSignalingService
participant VideoRoom as Live Interview Room (WebRTC Mesh)
participant DB as Database

HR->>ATS: Create Job Opening & Set Criteria
Cand->>ATS: Submit Job Application & Upload Resume
HR->>ATS: Review & Move Candidate to 'INTERVIEW_ROUND_1'
HR->>ATS: Schedule Interview (Lead Interviewer + Panelists)
ATS->>DB: Generate Guest Access Token & Meeting Links
ATS-->>Cand: Send Automated Email with Guest Token URL
ATS-->>Panel: Send Calendar Invite with Session ID

par Room Entry
Cand->>VideoRoom: Join via /guest/:guestToken (Candidate Lobby)
Panel->>VideoRoom: Join via /interviews/:sessionId (Evaluator Portal)
end

VideoRoom->>Signal: Send 'JOIN' Signal { senderIdentity, role }
Signal-->>VideoRoom: Broadcast 'JOIN' to all room participants

rect rgb(240, 245, 255)
Note over VideoRoom, Signal: WebRTC Mesh with Deterministic Tie-Breaker
VideoRoom->>VideoRoom: Evaluate: if (myIdentity < peerId) generate OFFER
VideoRoom->>Signal: Targeted 'OFFER' -> remote peer
Signal-->>VideoRoom: Deliver 'OFFER'
VideoRoom->>Signal: Targeted 'ANSWER' -> offerer
VideoRoom->>Signal: Exchange Targeted 'ICE' Candidates
VideoRoom->>VideoRoom: P2P Audio / Video Connected (3-Way Split)
end

par Live Interview Interaction
Panel->>VideoRoom: In-Call Multi-Party Chat & Code Scratchpad
Cand->>VideoRoom: Video Response & In-Call Chat
Panel->>ATS: Save Live Evaluation Scorecard (Technical, Comm, Culture Fit)
end

Panel->>ATS: Finalize Interview (Advance to Round 2 or HIRE)
ATS->>DB: Update Stage to 'OFFER_EXTENDED'
HR->>ATS: Click 'Convert Candidate to Employee'
ATS->>DB: Provision Employee Record + Assign Quotas (Annual: 24, Sick: 12, Casual: 8)
ATS-->>Cand: Dispatched Formal Offer & Portal Onboarding Link
```

Flow 4: Time, Attendance & Shift Scheduling Workflow

Automates shift roster allocation, IP and geofenced clock-in validation, daily break tracking, late arrival detection, regularization workflows, and monthly attendance register compilation for payroll.

Operational Execution Steps:

- **1. Shift Assignment:** Employee is assigned to shift (e.g. General Shift GEN-01: 09:00 - 18:00).
- **2. Geofenced Clock-In:** Employee clocks in via web/mobile; system verifies GPS coordinates and office IP subnet.
- **3. Punch Categorization:** Punch before 09:15 is marked PRESENT; punch after 09:15 is flagged LATE_ARRIVAL.
- **4. Shift Duration & Breaks:** Tracks break intervals (lunch/tea); employee clocks out at shift completion.
- **5. Regularization Request:** If an anomaly occurs (missing punch/under-hours), employee submits regularization with reason.
- **6. Manager Sign-Off:** Reporting manager approves regularization; attendance record is locked for payroll calculation.

Mermaid Diagram Specification:

```
flowchart TD
Emp([Employee Arrives]) --> ClockInAction[Initiate Clock-In Web/Mobile]
ClockInAction --> SecurityCheck{Geofence & IP Whitelist Valid?}

SecurityCheck -->|Violated| FlagAnomaly[Flag Discrepancy & Alert Manager]
SecurityCheck -->|Valid| RecordPunch[Record Attendance Punch with Timestamp]

FlagAnomaly --> RecordPunch
RecordPunch --> ShiftMatch[Match against Shift Schedule GEN-01: 09:00 - 18:00]

ShiftMatch --> ComputeStatus{Punch Time vs Shift Start}
ComputeStatus -->|<= 09:15| StatusPresent[Status: PRESENT]
ComputeStatus -->|> 09:15| StatusLate[Status: LATE_ARRIVAL]

StatusPresent --> ActiveShift[Shift in Progress Track Lunch & Breaks]
StatusLate --> ActiveShift

ActiveShift --> ClockOutAction[Initiate Clock-Out]
ClockOutAction --> CalcDuration[Calculate Total Work Hours & Overtime]

CalcDuration --> CheckRegularization{Missing Punch / Under Hours?}
CheckRegularization -->|Yes| ApplyReg[Submit Regularization Request]
ApplyReg --> MgrApproval{Manager Approves?}
MgrApproval -->|Yes| UpdatePresent[Update Register to REGULARIZED_PRESENT]
MgrApproval -->|No| KeepDeduction[Maintain Deduction / Half-Day]
CheckRegularization -->|No| RegisterClean[Attendance Finalized for Monthly Payroll]
UpdatePresent --> RegisterClean
KeepDeduction --> RegisterClean
```

Flow 5: Leave Management & Statutory Accrual Ledger

Controls statutory and corporate leave policies, annual allocation, monthly accruals, balance deduction ledgers, multi-tier approval chains, and attendance synchronization.

Operational Execution Steps:

- **1. Policy Setup:** HR configures leave categories (Annual, Sick, Casual, Maternity) with accrual limits.
- **2. Quota Ledgering:** Employee leave balances are provisioned and stored in EmployeeLeaveBalance entity.
- **3. Request Application:** Employee submits leave dates and reason in self-service portal.
- **4. Rule Validation:** System checks sufficient remaining balance and verifies no overlapping leave bookings.
- **5. Manager Approval Chain:** Notification routed to manager; manager reviews team calendar and approves or rejects.
- **6. Ledger Audit Deduction:** Units deducted from balance; audit record logged in LeaveLedger; syncs with payroll engine.

Mermaid Diagram Specification:

```
flowchart TD
Admin([HR Admin]) --> ConfigPolicy[Define Leave Policies Annual, Sick, Casual]
ConfigPolicy --> InitBalances[Allocate Employee Annual Balances]

Emp([Employee]) --> ViewBalance[View Available Quotas in Self-Service]
ViewBalance --> SubmitLeave[Submit Leave Request Dates, Reason, Type]

SubmitLeave --> ValidateRule{Sufficient Balance & No Date Conflict?}
ValidateRule -->|No| RejectInstant[Immediate Rejection Insufficient Balance]
ValidateRule -->|Yes| RouteApprover[Route to Reporting Manager Inbox]

RouteApprover --> MgrDecision{Manager Decision}
MgrDecision -->|Reject| NotifyEmpReject[Notify Employee with Reason]
MgrDecision -->|Approve| DeductBalance[Deduct Units from EmployeeLeaveBalance]

DeductBalance --> RecordAudit[Create LeaveLedger Entry Audit Trail]
RecordAudit --> SyncCalendar[Sync Corporate Calendar & Attendance Register]
SyncCalendar --> PayrollSync[Flag as Paid/Unpaid Leave for Payroll Engine]
```

Flow 6: Payroll Processing & Statutory Tax Life Cycle

Executes monthly salary calculations by synchronizing attendance data, gross earnings, PF/ESI/TDS statutory deductions, payroll approval lock, PDF payslip generation, and bank payment export.

Operational Execution Steps:

- **1. Attendance & LOP Sync:** Aggregates finalized monthly attendance and unpaid leave days for all staff.
- **2. Component Calculation:** Loads SalaryStructure (Basic, HRA, Allowances) and computes Gross Earnings.
- **3. Statutory Deductions:** Calculates Provident Fund (12%), ESI, Professional Tax, and Tax Deducted at Source (TDS).
- **4. Net Pay Computation:** Net Salary = Gross Earnings - Total Deductions; generates preview record.
- **5. HR Audit & Lock:** HR audits variance reports and locks payroll run (Status = PAID).
- **6. Bank Transfer & Payslips:** Exports NEFT/RTGS bank transfer batch file; renders PDF payslips to employee portal.

Mermaid Diagram Specification:

```
flowchart TD
StartCycle([Initiate Monthly Payroll Run]) --> FetchAttendance[Aggregate Monthly Attendance Register]
FetchAttendance --> FetchLeave[Sync Approved Unpaid Leaves & LOP Days]

FetchLeave --> IterateEmployees[Iterate Active Employees for Tenant]
IterateEmployees --> LoadStructure[Load Salary Structure Basic, HRA, Allowances]

LoadStructure --> CalcGross[Calculate Gross Earnings = Base - LOP Deductions]
CalcGross --> CalcStatutory[Calculate PF 12%, ESI, Professional Tax & TDS]
CalcStatutory --> CalcNet[Net Pay = Gross Earnings - Total Deductions]

CalcNet --> GenerateRecord[Insert PayrollRecord & Payslip Entity]
GenerateRecord --> NextEmp{More Employees?}
NextEmp -->|Yes| IterateEmployees
NextEmp -->|No| HRReview[HR Preview: Variance & Total Outflow Analysis]

HRReview --> ApproveRun{HR Approves Payroll?}
ApproveRun -->|Needs Adjustment| AdjustComp[Adjust Ad-hoc Bonus / Deduction]
AdjustComp --> HRReview
ApproveRun -->|Approved| FinalizeRun[Lock Payroll Run Status = PAID]

FinalizeRun --> ExportBank[Export Bank NEFT/RTGS Batch Transfer File]
FinalizeRun --> PublishPayslips[Publish PDF Payslips to Employee Self-Service]
PublishPayslips --> NotifyEmployees[Dispatch Notification Alerts to Staff]
```

Flow 7: Performance Appraisal, OKR & 360-Degree Review

Coordinates corporate appraisal cycles, employee OKR goal setting, self-reviews, manager scoring, HR bell-curve calibration, and final increment/promotion recommendations.

Operational Execution Steps:

- **1. Cycle Setup:** HR Admin launches performance appraisal cycle (e.g. FY 2026 Annual Appraisal).
- **2. Goal / OKR Setting:** Employee defines quarterly goals with measurable KPIs; manager reviews and approves.
- **3. Self-Evaluation:** Employee completes self-review rating competencies (1-5 stars) and achievement notes.
- **4. Manager Review:** Reporting manager completes evaluation, adds qualitative remarks, and recommends promotion/increment.
- **5. HR Calibration:** HR runs normalization committee to align ratings with corporate bell-curve targets.
- **6. Score Release:** HR finalizes cycle; appraisal report, rating, and increment letter are published to employee.

Mermaid Diagram Specification:

```
sequenceDiagram
autonumber
actor HR as HR Admin
actor Emp as Employee
actor Mgr as Department Manager
participant Perf as Performance Service
participant DB as Database

HR->>Perf: Create Appraisal Cycle (e.g. FY 2026 Annual Review)
HR->>Perf: Define Rating Scales & Competencies
Emp->>Perf: Set Key Performance Indicators (KPIs) / OKRs
Mgr->>Perf: Review & Approve Employee OKRs

Note over Emp, Mgr: Mid-Cycle Check-in & Milestone Tracking

HR->>Perf: Open Self-Appraisal Window
Emp->>Perf: Submit Self-Review & Competency Ratings
HR->>Perf: Open Manager Evaluation Window
Mgr->>Perf: Submit Manager Review, Comments & Proposed Increment

HR->>Perf: Run HR Calibration & Normalization (Bell Curve Fit)
HR->>Perf: Finalize Appraisal Scores & Promotion List
Perf->>DB: Lock Appraisal Record (Status = COMPLETED)
Perf-->>Emp: Release Evaluation Report & Increment Letter
```

Flow 8: Employee Expense Reimbursement & Claim Workflow

Governs employee business expense claims, receipt ingestion, multi-tier manager and accounts verification, policy adherence checks, and payment reimbursement.

Operational Execution Steps:

- **1. Claim Creation:** Employee creates claim (Travel, Client Meals, Hardware) and enters amounts.
- **2. Proof Ingestion:** Uploads invoice receipts, tickets, or supporting bill documents.
- **3. Policy Verification:** System checks claim against category caps and policy rules; flags exceptions.
- **4. Manager Review:** Manager verifies business relevance and approves or returns for clarification.
- **5. Finance Audit:** Finance/Accounts desk audits tax numbers and payment receipts.
- **6. Settlement Disbursement:** Claim marked APPROVED_FOR_PAYMENT; disbursed via monthly payroll or direct bank transfer.

Mermaid Diagram Specification:

```
flowchart TD
Emp([Employee]) --> CreateClaim[Create Expense Claim Travel, Food, Supplies]
CreateClaim --> UploadReceipts[Upload Bills / Proof Documents]
UploadReceipts --> CheckLimits{Within Policy Limit & Category?}

CheckLimits -->|Exceeds Hard Limit| AlertWarning[Warning: Out-of-policy approval required]
CheckLimits -->|Within Limits| SubmitClaim[Submit Claim to Workflow]
AlertWarning --> SubmitClaim

SubmitClaim --> MgrApproval{Manager Approval}
MgrApproval -->|Rejected| RejectClaim[Return to Employee with Feedback]
MgrApproval -->|Approved| FinanceReview{Finance / Accounts Audit}

FinanceReview -->|Discrepancy| RequestClarification[Request Additional Receipts]
RequestClarification --> Emp
FinanceReview -->|Verified| ApproveFinance[Mark Claim as APPROVED_FOR_PAYMENT]

ApproveFinance --> SettlementChoice{Settlement Mode}
SettlementChoice -->|Payroll Add-on| AddToPayroll[Add to Next Month Payroll Run]
SettlementChoice -->|Direct Transfer| BankReimburse[Direct Account Reimbursement]

AddToPayroll --> Settled([Claim Settled])
BankReimburse --> Settled
```

Flow 9: Company Hardware Asset Allocation & Tracking

Tracks company hardware assets (laptops, monitors, mobile devices), inventory status, employee allocation receipts, periodic health audits, and recovery upon offboarding.

Operational Execution Steps:

- **1. Inventory Ingestion:** IT logs new hardware with serial number, specs, and status AVAILABLE_IN_STOCK.
- **2. Requisition Submission:** Employee submits asset requisition for required equipment.
- **3. Manager Approval:** Manager approves necessity; requisition is dispatched to IT Asset Desk.
- **4. Asset Handover:** IT assigns specific asset (e.g. AST-LAP-001) and generates digital handover receipt.
- **5. Employee Acknowledgment:** Employee accepts condition; asset status transitions to ALLOCATED.
- **6. Deallocation & Return:** Upon exit or hardware refresh, asset is returned, inspected for damage, and restocked.

Mermaid Diagram Specification:

```
flowchart TD
IT([IT Admin]) --> OnboardAsset[Register Asset AST-LAP-001, AST-MON-002]
OnboardAsset --> SetInventory[Set Status = AVAILABLE_IN_STOCK]

Emp([Employee]) --> RequestAsset[Submit Asset Request Laptop / Monitor / Peripheral]
RequestAsset --> MgrApprove{Manager Approves Request?}
MgrApprove -->|No| DeclineReq[Decline Request]
MgrApprove -->|Yes| AssignAsset[IT Allocates Asset to Employee]

AssignAsset --> GenerateReceipt[Generate Asset Allocation Acknowledgment]
GenerateReceipt --> EmpSign[Employee Digitally Acknowledges Receipt]
EmpSign --> MarkAssigned[Set Asset Status = ALLOCATED]

MarkAssigned --> ActiveUsage[Active Usage & Periodic Health Audits]

TriggerReturn{Role Change / Exit Event?} --> ReturnAsset[Return Hardware to IT Desk]
ReturnAsset --> Inspection{Hardware Condition Check}
Inspection -->|Damaged| ChargeFee[Assess Deductible / Repair Fee]
Inspection -->|Good| Restock[Return to Inventory AVAILABLE_IN_STOCK]
```

Flow 10: Helpdesk & Employee Grievance Ticketing

Routes employee inquiries, HR complaints, IT requests, and payroll discrepancies with automated SLA tracking, agent assignment, and resolution sign-off.

Operational Execution Steps:

- **1. Ticket Filing:** Employee files ticket with category (Payroll, HR Policy, IT), priority, and description.
- **2. SLA & Queue Routing:** System computes SLA resolution deadline and assigns ticket to available agent.
- **3. Agent In-Progress:** Agent accepts ticket; updates status to IN_PROGRESS and requests clarifications if needed.
- **4. Communication Thread:** Two-way comments and document attachments between employee and agent.
- **5. Resolution Dispatch:** Agent documents solution and updates ticket status to RESOLVED.
- **6. Employee Closure:** Employee confirms resolution and provides Customer Satisfaction (CSAT) rating.

Mermaid Diagram Specification:

```
sequenceDiagram
autonumber
actor Emp as Employee
actor HR as HR / IT Helpdesk Agent
participant HD as Helpdesk Service
participant Notif as Notification Service

Emp->>HD: Create Ticket (Category: Payroll / IT / HR Policy, Priority: HIGH)
HD->>HD: Evaluate Category & SLA Deadline
HD->>Notif: Trigger Ticket Assignment Event
Notif-->>HR: Dispatch Alert to Assigned Agent
HR->>HD: Accept Ticket (Status = IN_PROGRESS)
HR->>HD: Post Clarification / Discussion Note
HD-->>Emp: Deliver In-App Update
Emp->>HD: Provide Supplementary Detail
HR->>HD: Resolve Ticket & Attach Documentation (Status = RESOLVED)
HD-->>Emp: Resolution Notification & Feedback Request
Emp->>HD: Close Ticket & Submit CSAT Rating
```

Flow 11: Employee Offboarding, Clearance & FnF Settlement

Orchestrates employee resignation, notice period administration, 4-desk clearances (IT, Finance, Admin, HR), asset returns, and Full & Final (FnF) financial settlement.

Operational Execution Steps:

- **1. Resignation Filing:** Employee submits resignation in self-service specifying reason and target Last Working Day (LWD).
- **2. Manager Acceptance:** Manager reviews notice period requirement and confirms official Last Working Day.
- **3. Clearance Checklist:** System automatically initiates clearance requests across IT, Finance, Admin, and Project Manager.
- **4. Multi-Desk Sign-off:** IT recovers assets, Finance audits loans, Admin reclaims ID badges, Manager verifies KT.
- **5. FnF Calculation:** Settlement engine computes: Unpaid Salary + Encashable Leaves - Shortfall Deductions.
- **6. Exit Closure:** Finance disburses final payment; HR issues Relieving Letter; account is archived.

Mermaid Diagram Specification:

```
flowchart TD
Emp([Employee]) --> Resign[Submit Resignation with Reason & Target LWD]
Resign --> MgrReview{Manager Review}
MgrReview -->|Discuss & Retain| Retained[Resignation Withdrawn Retained]
MgrReview -->|Accept| AcceptExit[Accept Resignation & Confirm Last Working Day]

AcceptExit --> InitChecklist[Generate Multi-Department Clearance Checklist]

InitChecklist --> ITClearance[IT Desk: Recover Assets & Revoke Access]
InitChecklist --> FinanceClearance[Finance Desk: Verify Pending Loans & Expenses]
InitChecklist --> AdminClearance[Admin Desk: Return ID Cards & Access Badges]
InitChecklist --> ProjectClearance[Manager: Knowledge Transfer KT Sign-off]

ITClearance --> CheckAllCleared{All 4 Desks Approved?}
FinanceClearance --> CheckAllCleared
AdminClearance --> CheckAllCleared
ProjectClearance --> CheckAllCleared

CheckAllCleared -->|Pending| WaitClearing[Follow up on pending items]
WaitClearing --> CheckAllCleared
CheckAllCleared -->|Approved| CalcFnF[Run Full & Final FnF Settlement Engine]

CalcFnF --> FnFComponents[Calculate: Unpaid Salary + Leave Encashment - Notice Shortfall]
FnFComponents --> FinanceSignoff[Finance Disburses Final Settlement]
FinanceSignoff --> IssueDocs[Issue Relieving Letter & Experience Certificate]
IssueDocs --> ArchiveEmp[Set Employee Status = TERMINATED / EXIT]
```

Flow 12: Workforce Intelligence & AI Predictive Analytics

Continuously aggregates cross-functional HR telemetry to forecast headcount growth, compute flight/attrition risk scores, and monitor payroll expenditure variance.

Operational Execution Steps:

- **1. Telemetry Ingestion:** Aggregates compensation, tenure, attendance anomalies, appraisal ratings, and helpdesk tickets.
- **2. Analytical Aggregation:** Computes departmental headcounts, gender/grade diversity indices, and overtime costs.
- **3. Predictive Risk Scoring:** Applies heuristic attrition scoring model to classify employees (High / Medium / Low Flight Risk).
- **4. Cost Projection:** Forecasts next-quarter payroll liability and overtime expenditure.
- **5. Visual Dashboarding:** Presents real-time heatmaps, risk alerts, and trend lines in executive dashboards.
- **6. Board Export:** Generates exportable executive reports in PDF and CSV for leadership meetings.

Mermaid Diagram Specification:

```
flowchart TD
subgraph DataIngestion[Continuous Enterprise Telemetry Ingestion]
D1[Tenure & Compensation Data]
D2[Attendance & Overtime Patterns]
D3[Appraisal Ratings & Goal Velocity]
D4[Helpdesk Tickets & Sentiment]
end

subgraph AnalyticalEngine[Workforce Intelligence Engine]
Aggregate[Aggregate Metrics across Departments & Roles]
RiskModel[Compute Attrition Risk Score High / Med / Low]
CostForecast[Project Next Quarter Payroll Liability]
DiversityCalc[Calculate Gender, Grade & Location Diversity]
end

subgraph ExecutiveReporting[Actionable Executive Dashboards]
Viz1[Real-time Headcount & Growth Trends]
Viz2[Attrition Early-Warning Heatmap]
Viz3[Departmental Overtime & Attendance Variance]
Viz4[Exportable Board Presentation Reports CSV/PDF]
end

DataIngestion --> Aggregate
Aggregate --> RiskModel
Aggregate --> CostForecast
Aggregate --> DiversityCalc

RiskModel --> Viz2
CostForecast --> Viz4
Aggregate --> Viz1
Aggregate --> Viz3
```

3. Enterprise Operational Verification Checklist

Matrix verifying end-to-end functionality, database synchronization, API routing, and frontend rendering across all 16 core platform modules.

Module	Capabilities	Backend Service	Frontend View	Status
Authentication & RBAC	Multi-tenant login, JWT token pair (15m/7d), password reset, role permissions	AuthService, User, Role	LoginPage.jsx, ProtectedRoute.jsx	**Verified & Active**
Tenant & Org Setup	Multi-company branding, office locations, departments, designations	TenantService, DepartmentService	DepartmentPage.jsx, TenantPage.jsx	**Verified & Active**
Employee Lifecycle	Employee profiles, manager hierarchy, KYC documents, statutory data	EmployeeService, Employee	EmployeesPage.jsx, EmployeeDetailPage.jsx	**Verified & Active**
Recruitment & ATS	Job postings, applicant pipeline, candidate screening, offer letters	RecruitmentService, JobOpening, Candidate	RecruitmentPage.jsx, JobDetailPage.jsx	**Verified & Active**
Live Interview Room	3-way multi-peer WebRTC mesh, candidate spotlight, in-call chat, scorecard	InterviewSignalingService, InterviewSession	LiveInterviewRoomPage.jsx	**Verified & Active**
Attendance & Shifts	Shift rosters (GEN-01), geofenced clock-in, overtime, regularization	AttendanceService, Attendance, Shift	AttendancePage.jsx, ShiftPage.jsx	**Verified & Active**
Leave Management	Statutory quotas, ledger balances, approval chain, holiday sync	LeaveService, LeaveRequest, LeaveBalance	LeavePage.jsx, LeaveApprovalPage.jsx	**Verified & Active**
Payroll & Compensation	Salary component formula, monthly run, PDF payslips, tax regimes	PayrollService, SalaryStructure, Payslip	PayrollPage.jsx, PayslipViewModal.jsx	**Verified & Active**
Performance (PMS)	FY cycles, OKR setting, self-review, manager 360, calibration	PerformanceService, AppraisalCycle	PerformancePage.jsx, ReviewModal.jsx	**Verified & Active**
Company Assets	Asset inventory, allocation receipts, maintenance, recovery	AssetService, Asset, AssetAllocation	AssetManagementPage.jsx	**Verified & Active**
Expense Claims	Multi-category claims, bill uploads, approval routing, settlement	ExpenseService, ExpenseClaim	ExpensePage.jsx, ClaimDetailModal.jsx	**Verified & Active**
Helpdesk & Grievance	Ticket routing, priority SLA, resolution communication	HelpdeskService, HelpdeskTicket	HelpdeskPage.jsx, TicketThread.jsx	**Verified & Active**
Employee Offboarding	Resignation workflow, 4-desk clearances, asset return, FnF settle	OffboardingService, ExitClearance	OffboardingPage.jsx, FnFCalculator.jsx	**Verified & Active**
Workforce Intelligence	Attrition risk scoring, headcount analytics, payroll variance	WorkforceIntelligenceService	WorkforceIntelligencePage.jsx	**Verified & Active**
Document Vault	Encrypted storage, employee folders, compliance validity alerts	DocumentService, DocumentRecord	DocumentVaultPage.jsx	**Verified & Active**
Communications	Corporate announcements, policy articles, notification alerts	CommunicationService, Announcement	AnnouncementsPage.jsx, NoticeBar.jsx	**Verified & Active**